Akshay Tiwari

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Education

Shri Ramswaroop Memorial College of Engineering and Management — **Expected Graduation: 2028**
Bachelor of Technology in Computer Science and Engineering — *Lucknow, Uttar Pradesh*

Relevant Coursework

- Data Structures
- Object-Oriented Programming
- Operating Systems
- Computer Organization & Architecture
- Discrete Structures & Theory of Logic
- Probability and Statistics
- Deep Learning Fundamentals
- Machine Learning (via Andrew Ng Specialization)

Projects

Lumora — AI-Powered Study Assistant | *Python, Gemini, FAISS, Streamlit, LM Studio* — **June 2026**

- Developed a Retrieval-Augmented Generation (RAG) study assistant for AKTU semester subjects using semantic retrieval and vector search.
- Implemented heading-aware chunking, embedding generation, and FAISS-based indexing for academic question answering.
- Integrated Google Gemini with a local LLM fallback (Meta-Llama-3.1-8B via LM Studio) to improve system reliability.
- Designed confidence filtering and hallucination guard mechanisms, achieving approximately 90% retrieval reliability during evaluation.
- Built an interactive Streamlit interface with persistent chat history and diagram-aware retrieval support.

Cats vs Dogs Image Classifier | *Python, TensorFlow, Keras* — **November 2025**

- Developed a Convolutional Neural Network (CNN) for binary image classification of cats and dogs.
- Performed image preprocessing, augmentation, model training, and evaluation using TensorFlow and Keras.
- Built as part of the IBM AI Virtual Internship program to gain practical deep learning experience.

Sign Language Recognition using CNN | *Python, PyTorch, torchvision* — **December 2025**

- Developed a Convolutional Neural Network (CNN) using PyTorch to classify hand gesture alphabets from the Sign-MNIST dataset.
- Implemented custom Dataset and DataLoader pipelines for preprocessing and batch training of image data.
- Designed a multi-layer CNN architecture with Batch Normalization, Dropout, and MaxPooling for improved generalization.
- Applied data augmentation techniques including random rotation, cropping, flipping, and color jittering to enhance model robustness.

Technical Skills

Languages: Python, Java, SQL
Developer Tools: VS Code, Git, GitHub, Jupyter Notebook , LM Studio
Technologies/Frameworks: PyTorch, TensorFlow, Keras, FAISS, Sentence Transformers, Gemini API, Mistral OCR

Certifications

- Advanced Learning Algorithms — DeepLearning.AI
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI
- Getting Started with Deep Learning — NVIDIA
- IBM AI Virtual Internship (Sep 2025 – Nov 2025)
- Digital Application Fundamentals (STEM)